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MULTI-CORE FIBER AMPLIFIER AND OPTICAL AMPLIFICATION METHOD

Abstract

A multi-core fiber amplifier includes: a first optical multiplexer that couples excitation light with first signal light of a first wavelength band and second signal light of a second wavelength band, and outputs coupled light; a first amplifier that includes a rare-earth doped multi-core fiber, inputs output light from the first optical multiplexer, and amplifies and outputs the first signal light; a first optical demultiplexer that demultiplexes output light from the first amplifier into the first signal light, and the second signal light and the excitation light, and outputs demultiplexed light; a second amplifier that includes a rare-earth doped multi-core fiber, inputs the second signal light and the excitation light from the first optical demultiplexer, and amplifies and outputs the input second signal light; and a second optical multiplexer that multiplexes the first signal light from the first optical demultiplexer and the second signal light from the second amplifier.

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Background/Summary

[0001] This application is based upon and claims the benefit of priority from Japanese patent application No. 2024-030933, filed on Mar. 1, 2024, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a multi-core fiber amplifier and an optical amplification method.

BACKGROUND ART

[0003] A space division multiplexing system using a multi-core fiber (MCF) is proceeding toward practical utilization. The MCF includes a plurality of cores in one optical fiber. Thus, an increase in transmission capacity of an optical transmission system can be achieved by using the MCF instead of a single core fiber (SCF). Further, a multi-core fiber amplifier (MCF amplifier) using the MCF as an excitation fiber is also known. An MCF in which erbium (Er) is added to a core is also referred to as a multi-core erbium-doped fiber (MC-EDF).

[0004] Wavelength division multiplexing signal light is usually used as signal light to be transmitted by the MCF. As a band of the wavelength division multiplexing signal light, an S-band, a C-band, and an L-band are known. A band of the S-band is approximately 1460 to 1530 nm, a band of the C-band is approximately 1530 to 1565 nm, and a band of the L-band is approximately 1565 to 1625 nm. The wavelength division multiplexing signal light is also referred to as wavelength division multiplexing (WDM) light.

[0005] In relation to the present disclosure, Patent Literature 1 (PTL 1) (International Patent Publication No. WO2022/181294) describes a multi-core EDFA (MC-EDFA) in which MC-EDFs are arranged in parallel. EDFA is an abbreviation for an Er-doped fiber amplifier. FIG. 10 is a diagram illustrating a configuration of a multi-core fiber amplifier 900 described in PTL 1. The multi-core fiber amplifier 900 includes multi-core EDFs 901 and 902, and amplifies, for each wavelength band, signal light in the C-band and signal light in the L-band that are output from a signal source 910. An optical demultiplexer 911 demultiplexes the signal light (C+L) being input from an MCF transmission path into the signal light (C) in the C-band and the signal light (L) in the L-band. The signal light in the C-band is multiplexed with excitation light (p) being output from an excitation light source 950 by an optical multiplexer 921, and multiplexed light is amplified in a multi-core EDF 901. An optical demultiplexer 912 demultiplexes the signal light in the C-band and the excitation light (residual excitation light) that are output from the multi-core EDF 901. The demultiplexed residual excitation light is multiplexed with the signal light in the L-band in an optical multiplexer 923, and multiplexed light is input to a multi-core EDF 902. The multi-core EDF 902 amplifies the signal light in the L-band by the residual excitation light. The residual excitation light is absorbed in the multi-core EDF 902. An optical multiplexer 922 multiplexes the amplified signal light in the C-band and the amplified signal light in the L-band, and outputs multiplexed light to an MCF transmission path 932.

SUMMARY

[0006] The multi-core fiber amplifier 900 illustrated in FIG. 10 excites the multi-core EDF 902 by using the residual excitation light being output from the multi-core EDF 901. Thus, the multi-core fiber amplifier 900 can excite the two MC-EDFs 901 and 902 disposed in parallel by using one excitation light source.

[0007] However, in the multi-core fiber amplifier 900, the optical demultiplexer 912 and the optical

demultiplexer **923** respectively demultiplexing and multiplexing the residual excitation light are essential on an output side of the multi-core EDF **901** and on an input side of the multi-core EDF **902** for supplying the residual excitation light to the multi-core EDF **902**. As a result, there is a problem that a loss of the excitation light and the signal light passing through the optical demultiplexer **912** and the optical multiplexer **923** increases and efficiency (amplification efficiency or power efficiency) of the multi-core fiber amplifier **900** decreases.

[0008] The present disclosure provides a technique for improving efficiency of a multi-core optical amplifier that amplifies light beams in a plurality of wavelength bands by using an excitation light source having a single wavelength.

[0009] A multi-core fiber amplifier according to the present disclosure includes: [0010] a first optical multiplexing means for coupling excitation light having a single wavelength and being used for amplification of signal light with first signal light being the signal light in a first wavelength band and second signal light being the signal light in a second wavelength band, and outputting coupled light; [0011] a first amplification means including a multi-core fiber in which a rare-earth element is added, for inputting output light from the first optical multiplexing means, and amplifying and outputting at least the first signal light; [0012] a first optical demultiplexing means for demultiplexing output light from the first amplification means into the first signal light, and the second signal light and the excitation light, and outputting demultiplexed light; [0013] a second amplification means including a multi-core fiber in which a rare-earth element is added, for inputting the second signal light and the excitation light that are output from the first optical demultiplexing means, and amplifying and outputting the input second signal light; and [0014] a second optical multiplexing means for multiplexing the first signal light being output from the first optical demultiplexing means and the second signal light being output from the second amplification means.

[0015] An optical amplification method according to the present disclosure includes: [0016] by a first optical multiplexing means, coupling excitation light having a single wavelength and being used for amplification of signal light with first signal light being the signal light in a first wavelength band and second signal light being the signal light in a second wavelength band, and outputting coupled light; [0017] inputting output light from the first optical multiplexing means to a first amplification means including a multi-core fiber in which a rare-earth element is added, and amplifying and outputting at least the first signal light; [0018] by a first optical demultiplexing means, demultiplexing output light from the first amplification means into the first signal light, and the second signal light and the excitation light, and outputting demultiplexed light; [0019] inputting the second signal light and the excitation light that are output from the first optical demultiplexing means to a second amplification means including a multi-core fiber in which a rare-earth element is added, and amplifying and outputting the input second signal light; and, [0020] by a second optical multiplexing means, multiplexing the first signal light being output from the first optical demultiplexing means and the second signal light being output from the second amplification means.

[0021] A multi-core fiber amplifier and the like according to the present disclosure can improve efficiency of a multi-core optical amplifier that amplifies light beams in a plurality of wavelength bands by using an excitation light source having a single wavelength.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] Exemplary features and advantages of the present invention will become apparent from the following detailed description when taken with the accompanying drawings in which:

[0023] FIG. **1** is a diagram illustrating a configuration example of a multi-core fiber amplifier;

[0024] FIG. 2 is a diagram illustrating a configuration example of the multi-core fiber amplifier;
[0025] FIG. 3 is a diagram illustrating a configuration example of an optical demultiplexer and an excitation light adjuster;
[0026] FIG. 4 is a diagram illustrating a configuration example of the optical demultiplexer and the excitation light adjuster;
[0027] FIG. 5 is a diagram illustrating a configuration example of the multi-core fiber amplifier;
[0028] FIG. 6 is a diagram illustrating a configuration example of the multi-core fiber amplifier;
[0029] FIG. 7 is a diagram illustrating a configuration example of the multi-core fiber amplifier;
[0030] FIG. 8 is a diagram illustrating a configuration example of the multi-core fiber amplifier;
[0031] FIG. 9 is a flowchart illustrating an example of an optical amplification method; and
[0032] FIG. 10 is a diagram illustrating a configuration of a multi-core fiber amplifier described in PTL 1.

EXAMPLE EMBODIMENT

[0033] Hereinafter, example embodiments of the present disclosure will be described with reference to drawings. An arrow illustrated in the drawings indicates a direction of a signal and the like, and does not intend to impose limitations on properties such as a direction and a size of a signal and the like. Further, an intersection point of paths indicated by a solid line does not mean coupling of the paths unless otherwise specified. A component that has already been described may be provided with the same reference sign in the example embodiments and the drawings, and redundant description thereof may be omitted.

First Example Embodiment

[0034] FIG. 1 is a diagram illustrating a configuration example of a multi-core fiber amplifier **100**. The multi-core fiber amplifier **100** includes multi-core EDFs **101** and **102**, optical multiplexers **111** and **112**, an optical demultiplexer **121**, and an excitation light source **150**. MCF transmission paths **161** and **162** are connected to the multi-core fiber amplifier **100**. The MCF transmission paths **161** and **162** are an optical transmission path formed of an MCF. The multi-core fiber amplifier **100** amplifies signal light being input from each core of the MCF transmission path **161**, and outputs the signal light to the MCF transmission path **162**.

[0035] The multi-core erbium-doped fibers (MC-EDFs) **101** and **102** are both an amplification medium, and amplify input light by excitation light. The multi-core EDFs **101** and **102** are an EDF in which erbium being a rare-earth element is added to each core of a multi-core fiber. The multi-core EDFs **101** and **102** can be referred to as a first amplification means and a second amplification means, respectively.

[0036] The excitation light and the signal light are input to the optical multiplexer **111**. Light having a single wavelength in a band of 0.98 μm and being generated in the excitation light source **150** can be used as the excitation light. The signal light is generated in a signal source **160** such as an optical transmitter, and propagates through the MCF transmission path **161**. The signal light is light in which first signal light being light in a first wavelength band and second signal light being light in a second wavelength band without overlapping the first wavelength band are wavelength-multiplexed. The optical multiplexer **111** couples the excitation light and the signal light, and inputs coupled light to the multi-core EDF **101**. The multi-core EDF **101** is mainly used for amplifying the first signal light. Further, the multi-core EDF **102** is entirely used for amplifying the second signal light. In the present example embodiment and subsequent example embodiments, the first wavelength band can be set as a C-band, and the second wavelength band can be set as an L-band. Note that clad excitation can be used as an excitation method of the multi-core EDFs **101** and **102**.

[0037] Note that FIG. 1 and subsequent diagrams appropriately illustrate graphs clearly illustrating a wavelength and intensity of signal light by indicating the first wavelength band as “1”, the second wavelength band as “2”, a wavelength in a horizontal axis, and light intensity in a vertical axis. However, in each of the graphs, light intensity in the vertical axis indicates a relative intensity difference between wavelength bands (for example, the C-band and the L-band) in the same graph,

and does not strictly indicate a difference in light intensity between different graphs.

[0038] The optical demultiplexer **121** is disposed between the multi-core EDF **101** and the multi-core EDF **102**. The optical demultiplexer **121** demultiplexes light output from the multi-core EDF **101** according to a wavelength band. One of the demultiplexed light beams includes the excitation light and the second signal light. The other demultiplexed light beam includes the first signal light. One of the demultiplexed light beams is output to the multi-core EDF **102**. The other demultiplexed light beam is output to the optical multiplexer **112**. As the optical demultiplexer **121**, an optical filter that passes the excitation light and the second signal light and reflects the first signal light can be used. The optical filter may be an optical band-pass filter. In this case, a wavelength band passed through the optical filter and a wavelength band reflected by the optical filter may be opposite.

[0039] The optical multiplexer **112** multiplexes the other light beam (the first signal light) demultiplexed in the optical demultiplexer **121** and the second signal light output from the multi-core EDF **102**, and outputs coupled light.

[0040] The multi-core fiber amplifier **100** having such a configuration can improve efficiency of a multi-core optical amplifier that amplifies light in a plurality of wavelength bands (i.e., a plurality of frequency bands) by using the excitation light source **150** having a single wavelength. The reason is described as follows.

[0041] In the multi-core fiber amplifier **100**, the multi-core EDF **101** and the multi-core EDF **102** are connected in series via the optical demultiplexer **121**. Thus, the multi-core EDFs **101** and **102** can be excited by using only excitation light having a single wavelength and being output from one excitation light source. With this configuration, an optical part (for example, the optical demultiplexer **912** and the optical multiplexer **923** illustrated in FIG. **10**) for injecting excitation light into the multi-core EDF **102** is unnecessary, and amplification efficiency of the multi-core EDF **102** can be improved. Further, since the first signal light amplified in the multi-core EDF **101** is separated in the optical demultiplexer **121**, the first signal light is output from the multi-core fiber amplifier **100** without suffering an unnecessary loss in the multi-core EDF **102**. Therefore, the multi-core fiber amplifier **100** can improve power conversion efficiency (PCE) as the entire optical amplifier when the first signal light and the second signal light are amplified.

[0042] Note that an absorption rate of the excitation light in the multi-core EDF **101** may be equal to or less than an absorption rate of the excitation light in the multi-core EDF **102**. By disposing an MC-EDF having a relatively low absorption rate of the excitation light on a prior stage, excitation light power being input to an MC-EDF on a subsequent stage can be increased, and thus an operation condition of the MC-EDF on the subsequent stage can be expanded.

Other Configuration of Multi-core Fiber Amplifier **100**

[0043] The effects of the multi-core fiber amplifier **100** described by using FIG. **1** can also be acquired by the following configuration. A reference sign in parentheses is the reference sign in FIG. **1**. The multi-core fiber amplifier **100** includes a first amplification means (**101**), a second amplification means (**102**), a first optical multiplexing means (**111**), a first optical demultiplexing means (**121**), and a second optical demultiplexing means (**112**).

[0044] The first amplification means and the second amplification means both include a multi-core fiber in which a rare-earth element is added. The first optical multiplexing means couples excitation light to first signal light and second signal light, and outputs coupled light. Output light from the first optical multiplexing means is input to the first amplification means. The first amplification means inputs the output light from the first optical multiplexing means, and amplifies and outputs at least the first signal light. The first optical demultiplexing means demultiplexes the light output from the first amplification means into the “first signal light” and the “second signal light and the excitation light”. The second signal light and the excitation light demultiplexed in the first optical demultiplexing means are input to the second amplification means. The second amplification means inputs the second signal light and the excitation light that are output from the first optical demultiplexing means, and amplifies and outputs the input second signal light. The

second optical multiplexing means multiplexes the first signal light demultiplexed in the first optical demultiplexing means and the second signal light output from the second amplification means. Note that the signal source **160** and the excitation light source (excitation light generation means) **150** are not essential in this configuration.

[0045] FIG. **9** is a flowchart illustrating an example of an optical amplification method in the above-described configuration. In other words, FIG. **9** is one example of the optical amplification method used in the multi-core fiber amplifier including the first amplification means including the multi-core fiber and the second amplification means including the multi-core fiber. The optical amplification method includes the following procedures. The excitation light being light having a single wavelength is coupled to the first signal light and the second signal light (step **S01** in FIG. **9**), and coupled light is amplified in the first amplification means (step **S02**). Then, the light output from the first amplification means is demultiplexed into the “first signal light” and the “second signal light and the excitation light” (step **S03**). Furthermore, the demultiplexed second signal light is amplified by the second amplification means (step **S04**), and the first signal light demultiplexed in step **S03** and the second signal light output from the second amplification means are multiplexed (step **S05**).

Modification Example of First Example Embodiment

[0046] FIG. **2** is a block diagram illustrating a configuration example of a multi-core fiber amplifier **100A**. The multi-core fiber amplifier **100A** is different from the multi-core fiber amplifier **100** in a point that the multi-core fiber amplifier **100A** includes an excitation light adjuster **130**. The excitation light adjuster **130** is disposed between the optical demultiplexer **121** and the multi-core EDF **102**. The excitation light adjuster **130** is one example of an excitation light adjustment means for adjusting power of the excitation light input to the multi-core EDF **102**. The excitation light adjuster **130** and the optical demultiplexer **121** may be formed integrally.

[0047] The multi-core fiber amplifier **100A** can adjust excitation light intensity for each wavelength band of the signal light by including the excitation light adjuster **130**. For example, the multi-core fiber amplifier **100A** can amplify the second signal light in the multi-core EDF **102** under a more suitable condition by adjusting intensity of the excitation light being input to the multi-core EDF **102** by using the excitation light adjuster **130**.

[0048] FIG. **3** is a diagram illustrating a configuration example of the optical demultiplexer **121** and the excitation light adjuster **130**. The optical demultiplexer **121** includes an optical filter **171**. The optical filter **171** demultiplexes light input from the multi-core EDF **101** into the excitation light (in a band of 0.98 μm), light in the second wavelength band (L-band), and light in the first wavelength band (C-band). The excitation light and the second signal light are output to the multi-core EDF **102** via the excitation light adjuster **130**.

[0049] The excitation light adjuster **130** includes optical filters **131** and **132** and a variable optical attenuator (VOA) **133**. The optical filter **131** demultiplexes the light input from the optical demultiplexer **121** into the excitation light and the second signal light. The second signal light passes through the optical filters **131** and **132** and is output to the multi-core EDF **102**. The excitation light is output from the optical filter **131** to the variable optical attenuator **133**. The variable optical attenuator **133** attenuates power of the input excitation light. The excitation light output from the variable optical attenuator **133** is multiplexed to the second signal light in the optical filter **132**, and is input to the multi-core EDF **102**. An attenuation amount of the excitation light in the variable optical attenuator **133** may be controlled by an electric signal from the outside of the multi-core fiber amplifier **100A**.

[0050] FIG. **4** is a diagram illustrating a configuration example of the optical demultiplexer **121** and an excitation light adjuster **130A**. The excitation light adjuster **130A** can be used instead of the excitation light adjuster **130**. The excitation light adjuster **130A** includes an optical coupler **134** in addition to the optical filters **131** and **132** and the variable optical attenuator **133**. The optical coupler **134** couples the light input from the optical demultiplexer **121** and the excitation light. The

excitation light adjuster **130A** can increase power of the excitation light input to the multi-core EDF **102** by including the optical coupler **134**. For example, when the power of the excitation light that excites the multi-core EDF **102** is insufficient, the excitation light adjuster **130A** can increase the power of the excitation light input to the multi-core EDF **102** by adding excitation light from the outside. Note that the optical coupler **134** is, for example, an optical directional coupler or a polarization synthesis coupler, and can be referred to as a third optical multiplexing means. [0051] In this way, the excitation light adjusters **130** and **130A** can be used for adjusting the power of the excitation light output to the multi-core EDF connected to a subsequent stage thereof. Note that, also in the subsequent example embodiments, the power of the excitation light input to the MC-EDF disposed on a subsequent stage of the excitation light adjuster **130** can be adjusted by disposing the excitation light adjuster **130** between the MC-EDFs.

Second Example Embodiment

[0052] FIG. 5 is a diagram illustrating a configuration example of a multi-core fiber amplifier **200**. The multi-core fiber amplifier **200** is different from the multi-core fiber amplifier **100** in a point that the multi-core fiber amplifier **200** includes an optical demultiplexer **122** and an excitation light coupler **211**.

[0053] Excitation light and second signal light that are output from a multi-core EDF **102** are input to the optical demultiplexer **122**. The optical demultiplexer **122** demultiplexes the input light into the excitation light and the second signal light. The optical demultiplexer **122** is, for example, an optical filter. The second signal light demultiplexed in the optical demultiplexer **122** is output to an optical multiplexer **112**. The optical multiplexer **112** multiplexes first signal light output from an optical demultiplexer **121** and the second signal light output from the optical demultiplexer **122**, and outputs coupled light to an MCF transmission path **162**. Meanwhile, the excitation light demultiplexed in the optical demultiplexer **122** is output to the excitation light coupler **211**. The optical demultiplexer **122** can be referred to as a second optical demultiplexing means for demultiplexing the second signal light and the excitation light that are output from the multi-core EDF **102**, and inputting the demultiplexed second signal light to the second optical multiplexing means.

[0054] The excitation light coupler **211** couples the excitation light demultiplexed in the optical demultiplexer **122** and the excitation light output from an excitation light source **150**. The excitation light coupled in the excitation light coupler **211** is output to an optical multiplexer **111**. The excitation light coupler **211** is, for example, an optical directional coupler or a polarization synthesis coupler, which is not limited thereto. The optical multiplexer **111** multiplexes the excitation light coupled in the excitation light coupler **211** and the signal light, and outputs multiplexed light to a multi-core EDF **101**. The excitation light coupler **211** can be referred to as an excitation light coupling means for coupling the excitation light demultiplexed in the optical demultiplexer **122** and the excitation light output from the excitation light source **150**.

[0055] In the operation of the multi-core fiber amplifier **200**, the optical demultiplexer **122** and the excitation light coupler **211** are not essential. However, the multi-core fiber amplifier **200** can efficiently use the excitation light by including the optical demultiplexer **122** and the excitation light coupler **211**. The reason is that, by inputting the excitation light (residual excitation light) output from the multi-core EDF **102** to the multi-core EDF **101** again, the excitation light can be reused (recycled).

Third Example Embodiment

[0056] FIG. 6 is a diagram illustrating a configuration example of a multi-core fiber amplifier **300**. The multi-core fiber amplifier **300** is different from the multi-core fiber amplifier **100** in a point that the multi-core fiber amplifier **300** has a configuration for amplifying third signal light being signal light in a third wavelength band. The third wavelength band is, for example, an S-band. Specifically, the multi-core fiber amplifier **300** includes a multi-core EDF **103** and an optical demultiplexer **123**.

[0057] The multi-core EDF **103** and the optical demultiplexer **123** are disposed in series between an optical multiplexer **111** and a multi-core EDF **101**. The multi-core EDF **103** is disposed between the optical multiplexer **111** and the optical demultiplexer **123**, and the optical demultiplexer **123** is disposed between the multi-core EDF **103** and the multi-core EDF **101**.

[0058] Excitation light and signal light are input to the optical multiplexer **111**. The excitation light is, for example, light having a single wavelength in a band of 0.98 μm . In the present example embodiment, the signal light is light in which first signal light, second signal light, and the third signal light are wavelength-multiplexed. The optical multiplexer **111** couples the excitation light and the signal light, and inputs coupled light to the multi-core EDF **103**. The excitation light may be generated in an excitation light source **150**.

[0059] The multi-core EDF **103** is an amplification medium, and amplifies input light by the excitation light. The multi-core EDF **103** illustrated in FIG. **6** is an MC-EDF in which erbium is added to each core of a multi-core fiber, and is disposed on a prior stage of the multi-core EDF **101**. The multi-core EDF **103** can be referred to as a third amplification means having a function of amplifying the signal light by the excitation light. The multi-core EDF **103** mainly amplifies the third signal light. An absorption rate of the excitation light in the multi-core EDF **103** may be equal to or less than an absorption rate of the excitation light in the multi-core EDF **101**. By not setting an absorption rate of the excitation light of the MC-EDF (multi-core EDF **103**) on a prior stage high (for example, the absorption rate is set equal to or less than an absorption rate of the excitation light of the MC-EDF on a subsequent stage), excitation light power being input to the multi-core EDF **101** and a multi-core EDF **102** on a subsequent stage can be increased.

[0060] The optical demultiplexer **123** is disposed between the multi-core EDF **103** and the multi-core EDF **101**. The light output from the multi-core EDF **103** is input to the optical demultiplexer **123**. The optical demultiplexer **123** demultiplexes the input light into two light beams. One of the demultiplexed light beams includes the first signal light, the second signal light, and the excitation light. The other demultiplexed light beam includes the third signal light. One of the demultiplexed light beams is output to the multi-core EDF **101**. The other demultiplexed light beam is output to an optical multiplexer **112**. As the optical demultiplexer **123**, an optical filter that passes the excitation light, the first signal light, and the second signal light and reflects the third signal light can be used. The optical filter may be an optical band-pass filter. Further, a wavelength band passed through the optical filter and a wavelength band reflected by the optical filter may be opposite. The optical demultiplexer **123** can be referred to as a third optical demultiplexing means for demultiplexing the light output from the multi-core EDF **103** into the “light including the first signal light, the second signal light, and the excitation light” and the “third signal light included in the signal light”. Note that a configuration and a function of a subsequent stage of the multi-core EDF **101** are similar to those in the first example embodiment, and thus the description will be omitted.

[0061] In the present example embodiment, the optical multiplexer **112** multiplexes the third signal light output from the optical demultiplexer **123** in addition to the first signal light output from the optical demultiplexer **121** and the second signal light output from the multi-core EDF **102**. Then, the optical multiplexer **112** outputs multiplexed light to an MCF transmission path **162**.

[0062] The multi-core fiber amplifier **300** having such a configuration can improve efficiency of a multi-core optical amplifier that amplifies a plurality of signal light beams by using the excitation light source **150** having a single wavelength. The multi-core fiber amplifier **300** includes the three MC-EDFs, and can thus amplify signal light beams in three different wavelength bands by using the amplification medium having a suitable length for each of the signal light beams. Further, by setting each parameter (for example, each length) of the multi-core EDFs **101** to **103** to be suitable for amplification in a C-band, an L-band, and an S-band, the multi-core fiber amplifier **300** can efficiently amplify the signal light beams in the three wavelength bands by using one excitation light beam having a single wavelength.

First Modification Example of Third Example Embodiment

[0063] FIG. 7 is a diagram illustrating a configuration example of a multi-core fiber amplifier **300A**. The multi-core fiber amplifier **300A** is different from the multi-core fiber amplifier **300** illustrated in FIG. 6 in a point that the multi-core fiber amplifier **300A** includes excitation light adjusters **130** and **135**. A configuration of the excitation light adjuster **135** is basically similar to that of the excitation light adjuster **130**, and has a function of adjusting only power of the excitation light. However, in the excitation light adjuster **135**, a pass band of optical filters **131** and **132** is set in such a way that light in a first wavelength band and light in a second wavelength band propagate through an optical path without passing through a variable optical attenuator **133**. Further, in the multi-core fiber amplifier **300A**, the excitation light adjusters **130** and **135** may have the configuration of the excitation light adjuster **130A** that can add the excitation light.

[0064] The multi-core fiber amplifier **300A** having such a configuration can adjust excitation light power in an input of each of the multi-core

[0065] EDF **101** and the multi-core EDF **102**. As a result, the multi-core fiber amplifier **300A** can individually control each amplification property of the multi-core EDFs **102** and **103**.

Second Modification Example of Third Example Embodiment

[0066] FIG. 8 is a diagram illustrating a configuration example of a multi-core fiber amplifier **300B**. The multi-core fiber amplifier **300B** is different from the multi-core fiber amplifier **300** illustrated in FIG. 6 in a point that the multi-core fiber amplifier **300B** includes an optical demultiplexer **122** and an excitation light coupler **211**. A configuration of the optical demultiplexer **122** and the excitation light coupler **211** is similar to that in the multi-core fiber amplifier **200** described in FIG. 5. In other words, the multi-core fiber amplifier **300B** is acquired by applying the configuration of the multi-core fiber amplifier **200** to the multi-core fiber amplifier **300**.

[0067] In the multi-core fiber amplifier **300B**, the excitation light and the second signal light that are output from the multi-core EDF **102** are input to the optical demultiplexer **122**. The optical demultiplexer **122** demultiplexes the input light into the excitation light and the second signal light, and outputs the second signal light to the optical multiplexer **112**. Further, the optical demultiplexer **122** outputs the excitation light to the excitation light coupler **211**.

[0068] The excitation light coupler **211** couples the excitation light demultiplexed in the optical demultiplexer **122** and the excitation light output from the excitation light source **150**. The excitation light coupled in the excitation light coupler **211** is output to the optical multiplexer **111**. The optical multiplexer **111** multiplexes the excitation light coupled in the excitation light coupler **211** and the signal light, and outputs multiplexed light to the multi-core EDF **103**.

[0069] The multi-core fiber amplifier **300B** having such a configuration can efficiently use the excitation light, similarly to the multi-core fiber amplifier **200**. The reason is that, by inputting the excitation light output from the multi-core EDF **102** to the multi-core EDF **103** again, the excitation light can be reused (recycled).

[0070] Note that the configuration of the multi-core fiber amplifier **300A** and the multi-core fiber amplifier **300B** is not exclusive. In other words, the multi-core fiber amplifier **300** may have a configuration also including the optical demultiplexer **122** and the excitation light coupler **211** in FIG. 8 in addition to the excitation light adjusters **130** and **135** in FIG. 7. With such a configuration, the effects of both the multi-core fiber amplifiers **300A** and **300B** can be acquired.

[0071] Note that the example embodiments of the present disclosure may also be described as supplementary notes below, which is not limited thereto.

Supplementary Note 1

[0072] A multi-core fiber amplifier including: [0073] a first optical multiplexing means for coupling excitation light having a single wavelength and being used for amplification of signal light with first signal light being the signal light in a first wavelength band and second signal light being the signal light in a second wavelength band, and outputting coupled light; [0074] a first amplification means including a multi-core fiber in which a rare-earth element is added, for inputting output light from the first optical multiplexing means, and amplifying and outputting at

least the first signal light; [0075] a first optical demultiplexing means for demultiplexing output light from the first amplification means into the first signal light, and the second signal light and the excitation light, and outputting demultiplexed light; [0076] a second amplification means including a multi-core fiber in which a rare-earth element is added, for inputting the second signal light and the excitation light that are output from the first optical demultiplexing means, and amplifying and outputting the input second signal light; and [0077] a second optical multiplexing means for multiplexing the first signal light being output from the first optical demultiplexing means and the second signal light being output from the second amplification means.

Supplementary Note 2

[0078] The multi-core fiber amplifier according to supplementary note 1, wherein [0079] an absorption rate of the excitation light in the first amplification means is equal to or less than an absorption rate of the excitation light in the second amplification means.

Supplementary Note 3

[0080] The multi-core fiber amplifier according to supplementary note 1 or 2, wherein [0081] the first wavelength band is a C-band, and the second wavelength band is an L-band.

Supplementary Note 4

[0082] The multi-core fiber amplifier according to any one of supplementary notes 1 to 3, further including [0083] an excitation light adjustment means for adjusting intensity of the excitation light being input to the second amplification means.

Supplementary Note 5

[0084] The multi-core fiber amplifier according to supplementary note 4, wherein [0085] the excitation light adjustment means includes an optical attenuator that attenuates power of the excitation light.

Supplementary Note 6

[0086] The multi-core fiber amplifier according to supplementary note 4 or 5, wherein [0087] the excitation light adjustment means includes a third optical multiplexing means for inputting the excitation light to the second amplification means.

Supplementary Note 7

[0088] The multi-core fiber amplifier according to any one of supplementary notes 1 to 6, further including [0089] an excitation light generation means for generating the excitation light, and inputting the excitation light to the first optical multiplexing means.

Supplementary Note 8

[0090] The multi-core fiber amplifier according to supplementary note 7, further including: [0091] a second optical demultiplexing means for demultiplexing the second signal light and the excitation light that are output from the second amplification means, and inputting the demultiplexed second signal light to the second optical multiplexing means; and [0092] an excitation light coupling means for coupling the excitation light demultiplexed in the second optical demultiplexing means and the excitation light being output from the excitation light generation means.

Supplementary Note 9

[0093] The multi-core fiber amplifier according to any one of supplementary notes 1 to 8, further including: [0094] a third amplification means including a multi-core fiber in which a rare-earth element is added, for amplifying third signal light being the signal light in a third wavelength band; and [0095] a third optical demultiplexing means for demultiplexing light being output from the third amplification means into light including the first signal light, the second signal light, and the excitation light, and the third signal light, wherein [0096] the third amplification means and the third optical demultiplexing means are disposed between the first optical multiplexing means and the first amplification means, and [0097] the first optical multiplexing means couples the excitation light with the first signal light, the second signal light, and the third signal light, and outputs coupled light.

Supplementary Note 10

[0098] The multi-core fiber amplifier according to supplementary note 9, wherein [0099] an absorption rate of the excitation light in the third amplification means is equal to or less than an absorption rate of the excitation light in the first amplification means.

Supplementary Note 11

[0100] The multi-core fiber amplifier according to supplementary note 9 or 10, wherein [0101] the third wavelength band is an S-band.

Supplementary Note 12

[0102] An optical amplification method including: [0103] by a first optical multiplexing means, coupling excitation light having a single wavelength and being used for amplification of signal light with first signal light being the signal light in a first wavelength band and second signal light being the signal light in a second wavelength band, and outputting coupled light; [0104] inputting output light from the first optical multiplexing means to a first amplification means including a multi-core fiber in which a rare-earth element is added, and amplifying and outputting at least the first signal light; [0105] by a first optical demultiplexing means, demultiplexing output light from the first amplification means into the first signal light, and the second signal light and the excitation light, and outputting demultiplexed light; [0106] inputting the second signal light and the excitation light that are output from the first optical demultiplexing means to a second amplification means including a multi-core fiber in which a rare-earth element is added, and amplifying and outputting the input second signal light; and, [0107] by a second optical multiplexing means, multiplexing the first signal light being output from the first optical demultiplexing means and the second signal light being output from the second amplification means.

Supplementary Note 13

[0108] The optical amplification method according to supplementary note 12, wherein [0109] an absorption rate of the excitation light in the first amplification means is equal to or less than an absorption rate of the excitation light in the second amplification means.

Supplementary Note 14

[0110] The optical amplification method according to supplementary note 12 or 13, wherein [0111] the first wavelength band is a C-band, and the second wavelength band is an L-band.

Supplementary Note 15

[0112] The optical amplification method according to any one of supplementary notes 12 to 14, further including [0113] adjusting intensity of the excitation light being input to the second amplification means.

Supplementary Note 16

[0114] The optical amplification method according to supplementary note 15, further including [0115] attenuating power of the excitation light by an optical attenuator.

Supplementary Note 17

[0116] The optical amplification method according to supplementary note 15 or 16, further including [0117] inputting the excitation light to the second amplification means.

Supplementary Note 18

[0118] The optical amplification method according to any one of supplementary notes 12 to 17, further including [0119] generating the excitation light, and inputting the excitation light to the first optical multiplexing means.

Supplementary Note 19

[0120] The optical amplification method according to supplementary note 18, further including: [0121] by a second optical demultiplexing means, demultiplexing the second signal light and the excitation light that are output from the second amplification means; [0122] inputting the demultiplexed second signal light to the second optical multiplexing means; and [0123] coupling the demultiplexed excitation light and the excitation light.

Supplementary Note 20

[0124] The optical amplification method according to any one of supplementary notes 12 to 19,

further including: [0125] disposing a third amplification means including a multi-core fiber in which a rare-earth element is added, and a third optical demultiplexing means, between the first optical multiplexing means and the first amplification means; [0126] coupling, by the first optical multiplexing means, the excitation light with the first signal light, the second signal light, and third signal light being the signal light in a third wavelength band, and inputting coupled light to the third amplification means; [0127] amplifying, by the third amplification means, at least the third signal light being the signal light in the third wavelength band; and [0128] demultiplexing, by the third optical demultiplexing means, light being output from the third amplification means into light including the first signal light, the second signal light, and the excitation light, and the third signal light.

Supplementary Note 21

[0129] The optical amplification method according to supplementary note 20, wherein [0130] an absorption rate of the excitation light in the third amplification means is equal to or less than an absorption rate of the excitation light in the first amplification means.

Supplementary Note 22

[0131] The optical amplification method according to supplementary note 20 or 21, wherein [0132] the third wavelength band is an S-band.

[0133] While the invention has been shown and described with reference to exemplary embodiments thereof, the invention is not limited to these embodiments. It will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present invention as defined by the claims. For example, the description of the multi-core fiber amplifier in each of the example embodiments also discloses a configuration of an optical transmission system including the multi-core fiber amplifier and a signal source, an optical transmitter, an optical receiver, and an optical transceiver being connected to the multi-core fiber amplifier, and an optical amplification method in the multi-core fiber amplifier.

[0134] Further, the configuration described in each of the example embodiments is not necessarily exclusive. The action and effects of the present disclosure may be achieved by a configuration combining the whole or a part of the above-mentioned example embodiments.

Claims

1. A multi-core fiber amplifier comprising: a first optical multiplexer configured to couple excitation light having a single wavelength and being used for amplification of signal light with first signal light being the signal light in a first wavelength band and second signal light being the signal light in a second wavelength band, and output coupled light; a first amplifier configured to include a multi-core fiber in which a rare-earth element is added, input output light from the first optical multiplexer, and amplify and output at least the first signal light; a first optical demultiplexer configured to demultiplex output light from the first amplifier into the first signal light, and the second signal light and the excitation light, and output demultiplexed light; a second amplifier configured to include a multi-core fiber in which a rare-earth element is added, input the second signal light and the excitation light that are output from the first optical demultiplexer, and amplify and output the input second signal light; and a second optical multiplexer configured to multiplex the first signal light being output from the first optical demultiplexer and the second signal light being output from the second amplifier.

2. The multi-core fiber amplifier according to claim 1, wherein an absorption rate of the excitation light in the first amplifier is equal to or less than an absorption rate of the excitation light in the second amplifier.

3. The multi-core fiber amplifier according to claim 1, wherein the first wavelength band is a C-band, and the second wavelength band is an L-band.

4. The multi-core fiber amplifier according to claim 1, further comprising an excitation light adjuster configured to adjust intensity of the excitation light being input to the second amplifier.
5. The multi-core fiber amplifier according to claim 4, wherein the excitation light adjuster includes an optical attenuator that attenuates power of the excitation light.
6. The multi-core fiber amplifier according to claim 4, wherein the excitation light adjuster includes a third optical multiplexer configured to input the excitation light to the second amplifier.
7. The multi-core fiber amplifier according to claim 1, further comprising an excitation light generator configured to generate the excitation light, and input the excitation light to the first optical multiplexer.
8. The multi-core fiber amplifier according to claim 7, further comprising: a second optical demultiplexer configured to demultiplex the second signal light and the excitation light that are output from the second amplifier, and input the demultiplexed second signal light to the second optical multiplexer; and an excitation light coupler configured to couple the excitation light demultiplexed in the second optical demultiplexer and the excitation light being output from the excitation light generator.
9. The multi-core fiber amplifier according to claim 1, further comprising: a third amplifier configured to include a multi-core fiber in which a rare-earth element is added and amplify third signal light being the signal light in a third wavelength band; and a third optical demultiplexer configured to demultiplex light being output from the third amplifier into light including the first signal light, the second signal light, and the excitation light, and the third signal light, wherein the third amplifier and the third optical demultiplexer are disposed between the first optical multiplexer and the first amplifier, and the first optical multiplexer couples the excitation light with the first signal light, the second signal light, and the third signal light, and outputs coupled light.
10. The multi-core fiber amplifier according to claim 9, wherein an absorption rate of the excitation light in the third amplifier is equal to or less than an absorption rate of the excitation light in the first amplifier.
11. The multi-core fiber amplifier according to claim 9, wherein the third wavelength band is an S-band.
12. An optical amplification method comprising: by a first optical multiplexer, coupling excitation light having a single wavelength and being used for amplification of signal light with first signal light being the signal light in a first wavelength band and second signal light being the signal light in a second wavelength band, and outputting coupled light; inputting output light from the first optical multiplexer to a first amplifier including a multi-core fiber in which a rare-earth element is added, and amplifying and outputting at least the first signal light; by a first optical demultiplexer, demultiplexing output light from the first amplifier into the first signal light, and the second signal light and the excitation light, and outputting demultiplexed light; inputting the second signal light and the excitation light that are output from the first optical demultiplexer to second amplifier including a multi-core fiber in which a rare-earth element is added, and amplifying and outputting the input second signal light; and, by a second optical multiplexer, multiplexing the first signal light being output from the first optical demultiplexer and the second signal light being output from the second amplifier.
13. The optical amplification method according to claim 12, wherein an absorption rate of the excitation light in the first amplifier is equal to or less than an absorption rate of the excitation light in the second amplifier.
14. The optical amplification method according to claim 12, wherein the first wavelength band is a C-band, and the second wavelength band is an L-band.
15. The optical amplification method according to claim 12, further comprising adjusting intensity of the excitation light being input to the second amplifier.
16. The optical amplification method according to claim 15, further comprising attenuating power of the excitation light by an optical attenuator.

17. The optical amplification method according to claim 15, further comprising inputting the excitation light to the second amplifier.

18. The optical amplification method according to claim 12, further comprising generating the excitation light, and inputting the excitation light to the first optical multiplexer.

19. The optical amplification method according to claim 18, further comprising: by a second optical demultiplexer, demultiplexing the second signal light and the excitation light that are output from the second amplifier; inputting the demultiplexed second signal light to the second optical multiplexer; and coupling the demultiplexed excitation light and the excitation light.

20. The optical amplification method according to claim 12, further comprising: disposing a third amplifier including a multi-core fiber in which a rare-earth element is added, and third optical demultiplexer, between the first optical multiplexer and the first amplifier; coupling, by the first optical multiplexer, the excitation light with the first signal light, the second signal light, and third signal light being the signal light in a third wavelength band, and inputting coupled light to the third amplifier; amplifying, by the third amplifier, at least the third signal light being the signal light in the third wavelength band; and demultiplexing, by the third optical demultiplexer, light being output from the third amplifier into light including the first signal light, the second signal light, and the excitation light, and the third signal light.
